KCA UNIVERSITY | CPP 4201

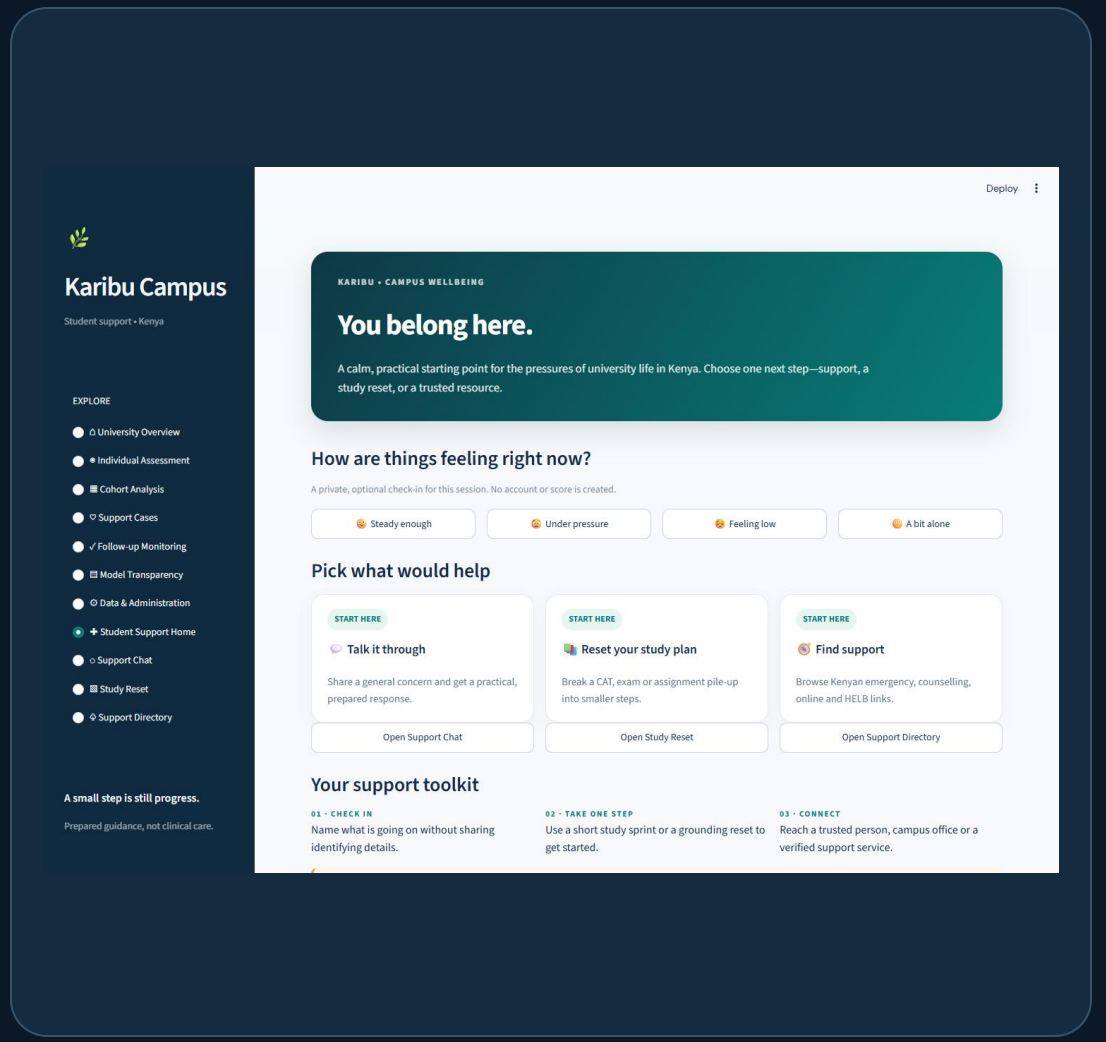
KARIBU CAMPUS

Student Support and Academic Outcome Prototype

Advanced Python Programming | Project Presentation

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karibu-campus.onrender.com



A Kenyan campus support interface with a separate outcome analytics pipeline

Student pressure and support access

University life can combine academic pressure with financial and social strain.

CAMPUS CHALLENGES

- CATs, exams, deadlines and missed coursework
- Fees, HELB/HEF uncertainty, upkeep, rent or fare
- Anxiety, isolation, low mood and difficulty finding help

TWO COMPLEMENTARY PROTOTYPE PATHS

Student support hub

Optional session check-in, prepared intent-based chat, study reset and a directory of human support routes.

Academic outcome analytics

Historical outcome analysis and synthetic-profile model assessment using the public UCI dataset.

The support chat and model remain separate; model scores never create support cases.

Two paths inside one Streamlit application

STUDENT SUPPORT PATH

Streamlit navigation
11 views in one prototype

→

Student support hub
Check-in, chatbot, planner, directory

↓

Human-owned cases
Pseudonymous local JSON demo

DATA AND MODEL PATH

Data preparation
UCI download, schema checks, split

→

Scikit-learn Pipeline
Fold-fitted encoding and scaling

↓

Model evaluation
CV comparison, holdout metrics, artifacts

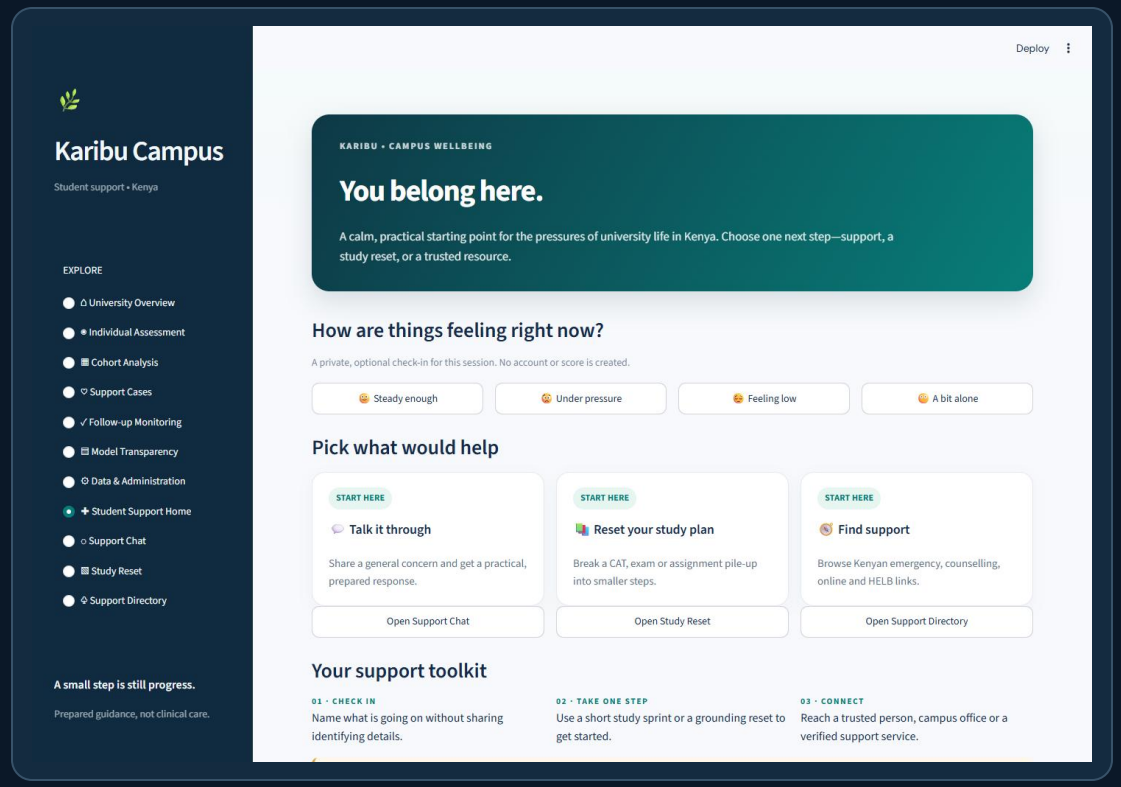
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Analytics views
Assessment, cohort, transparency

The support chat routes text to prepared replies. The modelling path reads the historical dataset and writes a local model plus reporting artifacts.

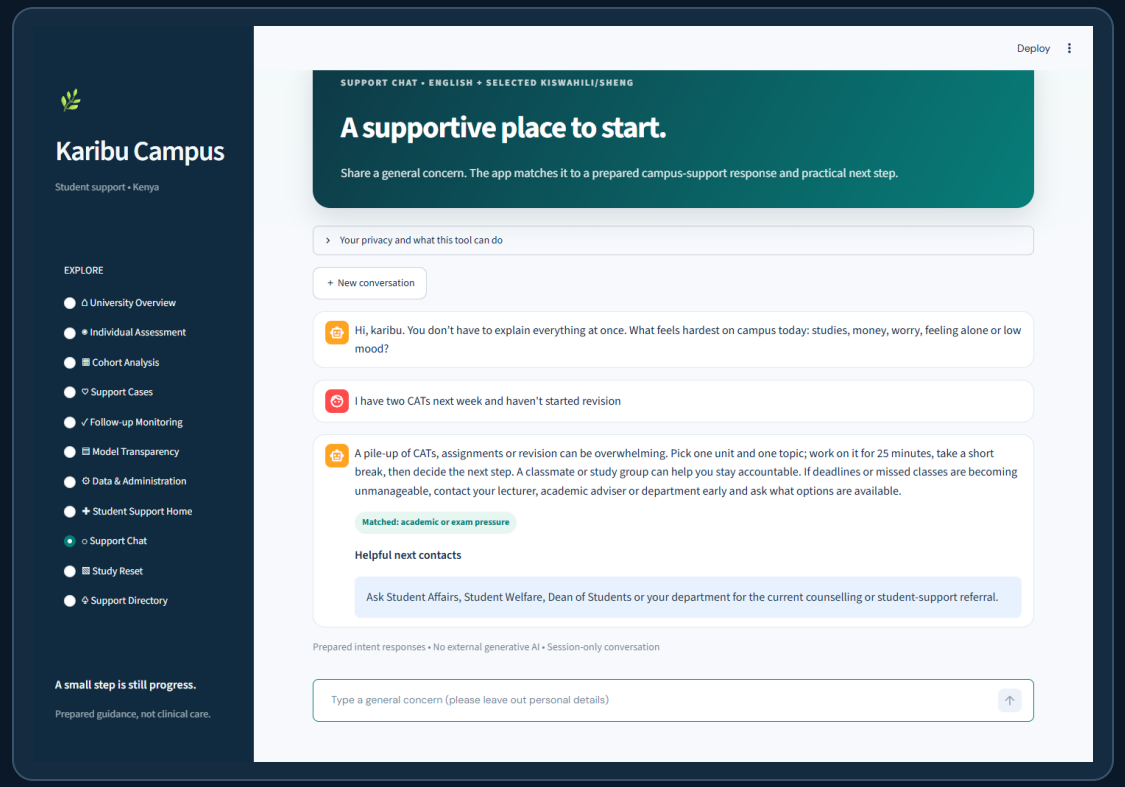
Student-facing navigation and intent-based chat

STUDENT SUPPORT HOME



Optional check-in stays in the current session; the page links to chat, study planning and resources.

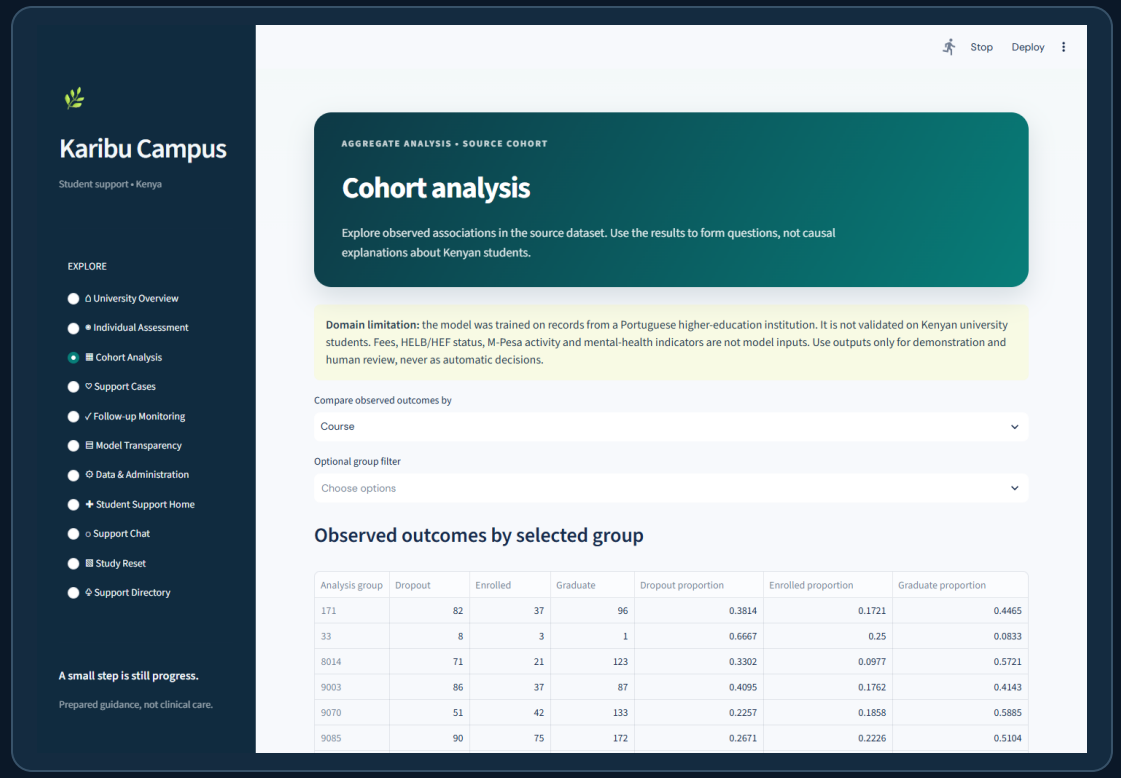
SUPPORT CHAT · CAT / REVISION EXAMPLE



Ordered regex patterns select a prepared reply and a human support next step.

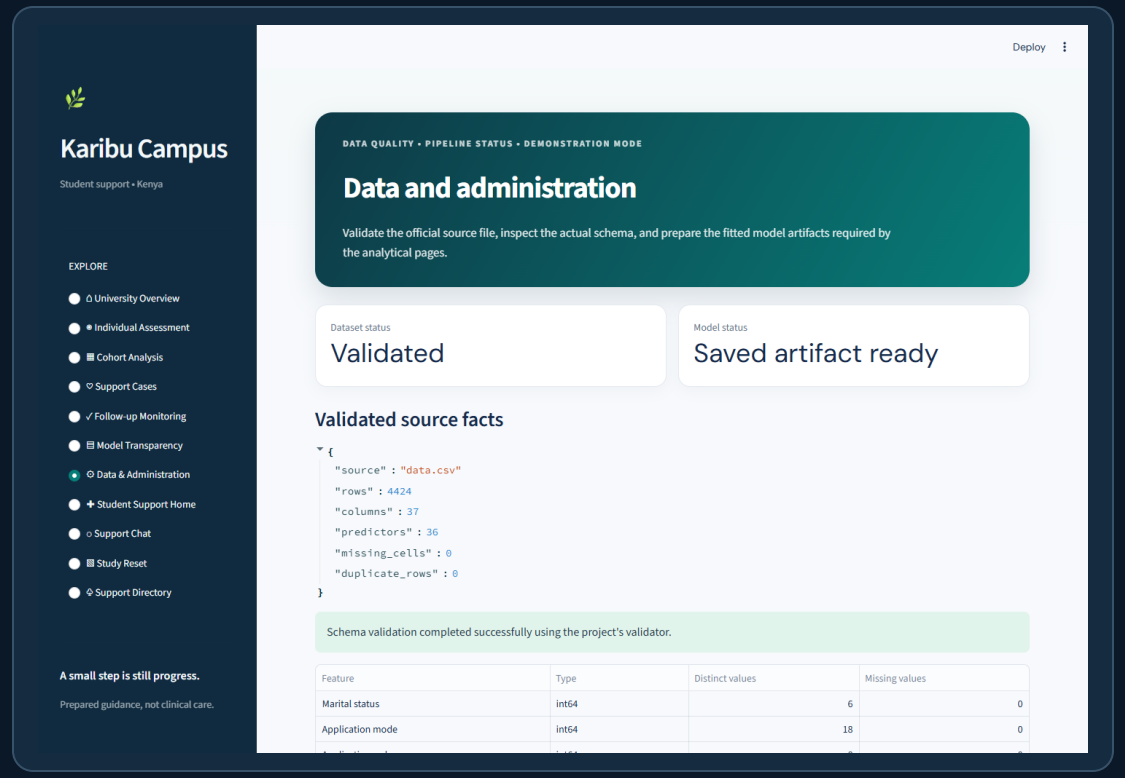
Administrative views support demonstration and review

COHORT ANALYSIS



Filters and grouped outcome counts describe the source cohort.

DATA & ADMINISTRATION



Schema checks and local training create artifacts for model-backed views.

Other views: Individual Assessment | Support Cases | Follow-up Monitoring | Model Transparency

UCI 697: source, structure and target

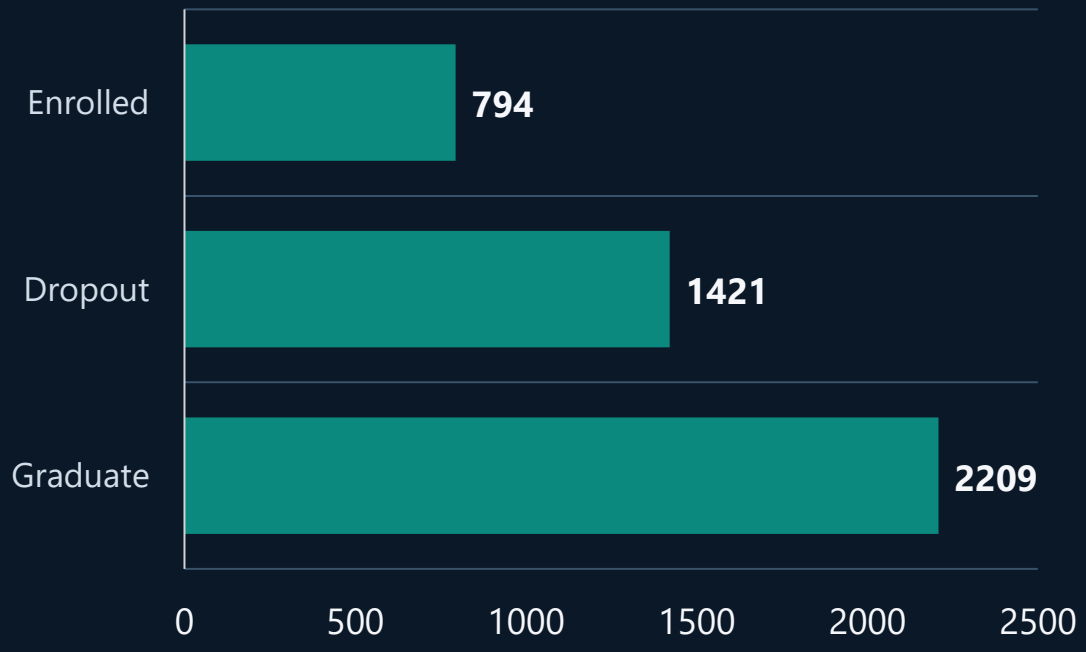
4,424 records
36 numeric predictors
1 observed outcome

Public UCI dataset 697, from a Portuguese higher-education institution. The CSV contains 37 columns and no missing cells in the inspected copy.

FEATURE GROUPS IN THE REPOSITORY

Student demographics and background
Admission, course and prior education
Fees, debtor and scholarship indicators
Semester enrolment, approvals, grades and evaluations
National economic indicators

OBSERVED TARGET CLASS COUNTS

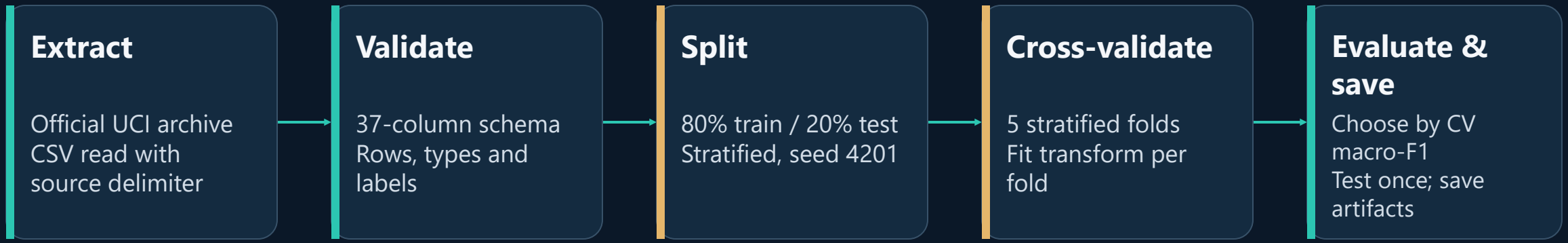


Enrolled 794 · Dropout 1,421 · Graduate 2,209

The Kenyan campus setting describes the support interface; it does not change the dataset's country of origin.

A reproducible pipeline with fold-fitted preprocessing

One stratified holdout for final evaluation; training folds for model selection.



INSIDE EACH MODEL PIPELINE

Numeric features are scaled. Nominal codes are one-hot encoded with unknown categories ignored. The transformed features then enter the classifier.

CANDIDATES

Logistic Regression
Random Forest
Gradient Boosting

SELECTION RULE

Highest mean training CV macro-F1. Logistic Regression won.

Held-out performance is uneven across outcome classes

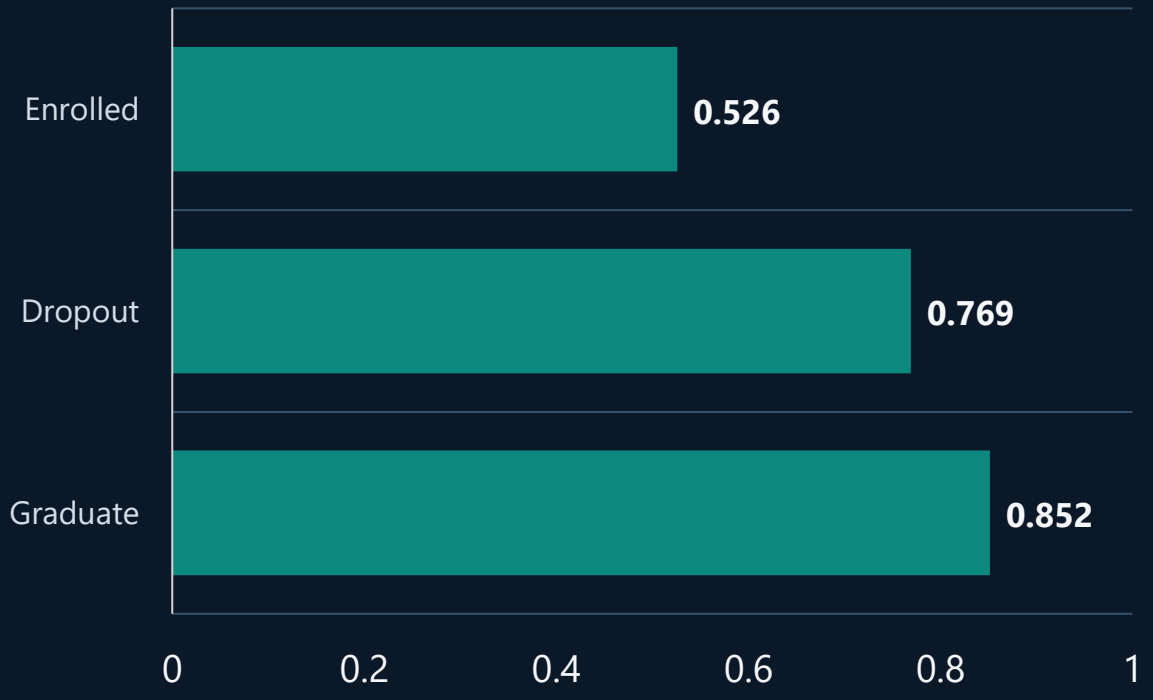
Selected model: Logistic Regression

Training CV macro-F1
0.705 ± 0.030

Held-out test set
885 records

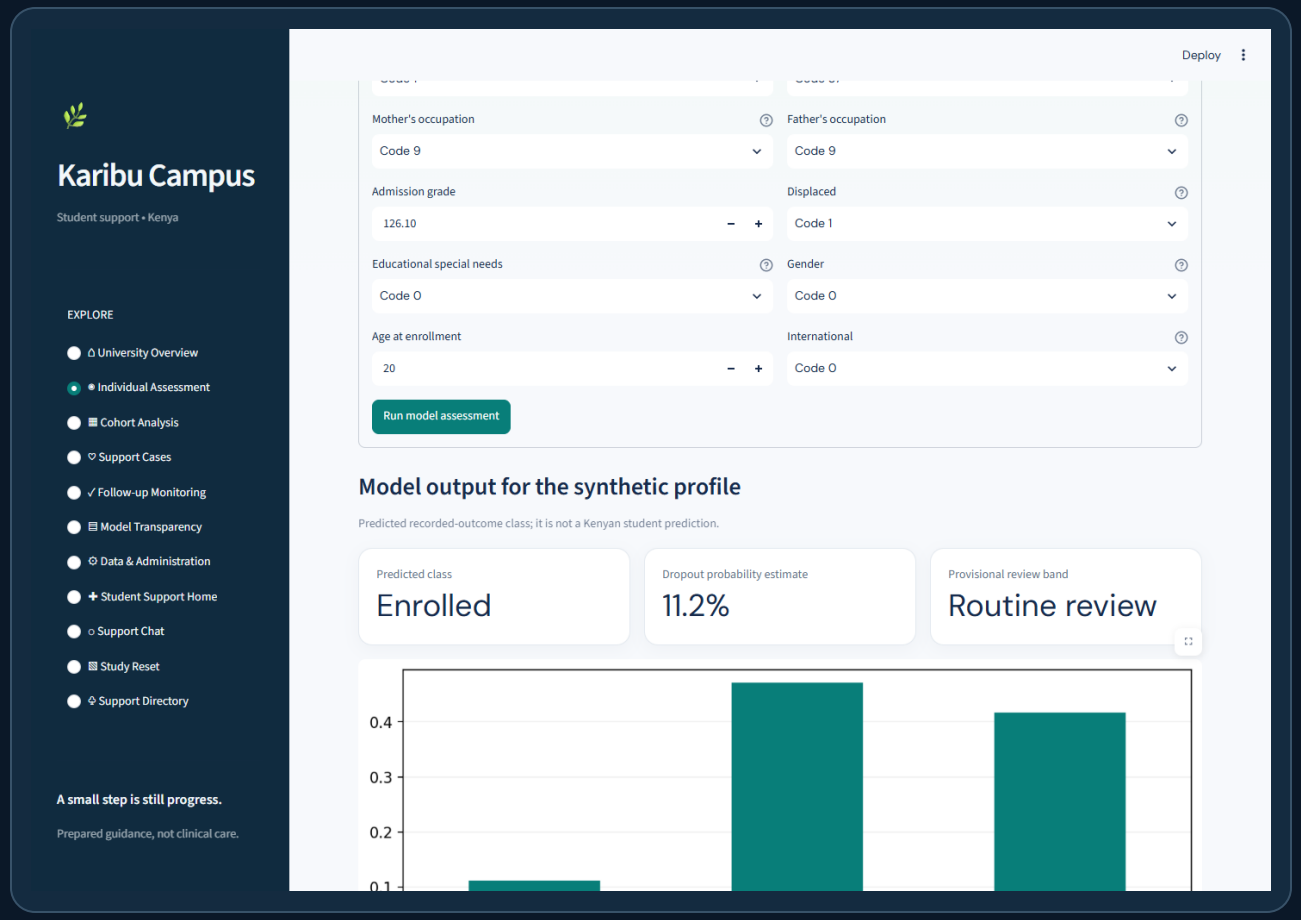
Macro-F1 0.716
Accuracy 0.756

PER-CLASS HELD-OUT F1



Enrolled F1 is 0.526; scores reflect Portuguese test data, not Kenya.

Assessment outputs invite review, not automated action



Repository screenshot · synthetic profile

WHAT THE PAGE SHOWS

- Three probabilities for the dataset's observed outcome classes
- A dropout probability and adjustable provisional review band
- Synthetic starting profile from medians and category modes

REVIEW BANDS

Routine below 0.35
Consider follow-up from 0.35
Priority human review from 0.65

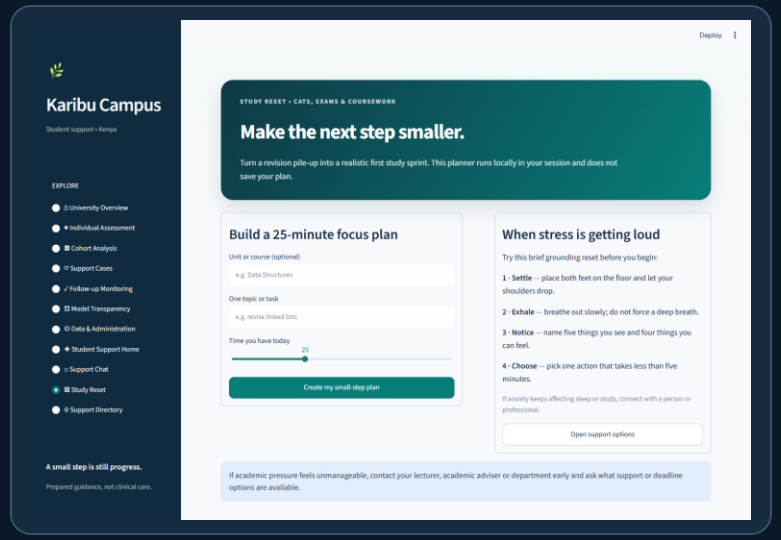
Probabilities are not calibrated confidence or causal explanations.

Limits that shape any responsible interpretation

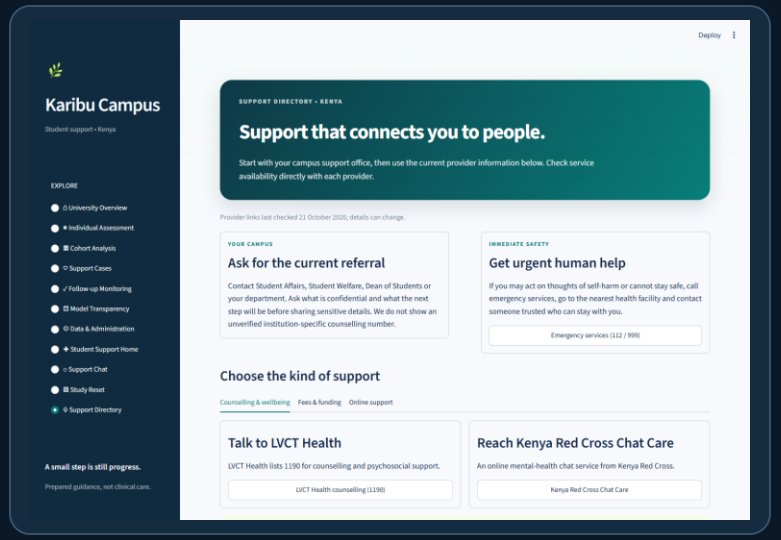
DATA TRANSFER	Portuguese institution; no external validation on Kenyan university students.
TIMING	Semester performance can be recorded close to the outcome, limiting early-intervention value.
CLASS PERFORMANCE	Enrolled has lower test F1. No subgroup fairness, calibration or prospective-impact study is reported.
CHAT SCOPE	Deterministic regex routing and prepared replies. It does not diagnose, monitor or understand every context.
CASE SECURITY	Pseudonymous local JSON demo without authentication, encryption, access controls or audit trail.
REFERRALS	Use current institutional routes and recheck provider details. The application is not a monitored emergency service.

A live tour links the student and analytics paths

STUDY RESET



SUPPORT DIRECTORY



DEMO SEQUENCE

1 Student Support Home
2 Support Chat: CAT or HELB example
3 Study Reset planner
4 Support Directory
5 Data & Administration
6 Prepare UCI data and train
7 Individual Assessment
8 Cohort Analysis

LIVE APP

https://karibu-campus.onrender.com

NEXT STEPS

Co-design with Kenyan students and support staff. Obtain governed local data, validate across groups and time, then add secure access and verified campus referrals.

For model pages, run the documented CLI to download, validate, train and create local artifacts before the live demo.

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Thank you

Questions & Discussion

A prototype can connect support access with outcome analysis. Responsible use still depends on Kenyan validation, privacy safeguards and meaningful human oversight.